

NILK: Entity Linking Dataset Targeting NIL-Linking Cases

Anastasiia Iurshina
anastasiia.iurshina@ipvs.uni-stuttgart.de
University of Stuttgart
Germany

Rafika Boutalbi
rafika.boutalbi@ipvs.uni-stuttgart.de
University of Stuttgart
Germany

Jiaxin Pan
jiaxin.pan@ipvs.uni-stuttgart.de
University of Stuttgart
Germany

Steffen Staab
steffen.staab@ipvs.uni-stuttgart.de
University of Stuttgart
Germany

ABSTRACT

The NIL-linking task in Entity Linking deals with cases where the text mentions do not have a corresponding entity in the associated knowledge base. NIL-linking has two sub-tasks: NIL-detection and NIL-disambiguation. NIL-detection identifies NIL-mentions in the text. Then, NIL-disambiguation determines if some NIL-mentions refer to the same out-of-knowledge base entity. Although multiple existing datasets can be adapted for NIL-detection, none of them address the problem of NIL-disambiguation. This paper presents NILK, a new dataset for NIL-linking processing, constructed from WikiData and Wikipedia dumps from two different timestamps. The NILK dataset has two main features: 1) It marks NIL-mentions for NIL-detection by extracting mentions which belong to newly added entities in Wikipedia text. 2) It provides an entity label for NIL-disambiguation by marking NIL-mentions with WikiData IDs from the newer dump. We make available the annotated dataset along with the code¹. The NILK dataset is available at: <https://zenodo.org/record/6607514>².

CCS CONCEPTS

• **Computing methodologies** → *Language resources*.

KEYWORDS

Datasets, Entity linking, Wikidata, NIL-mentions

ACM Reference Format:

Anastasiia Iurshina, Jiaxin Pan, Rafika Boutalbi, and Steffen Staab. 2022. NILK: Entity Linking Dataset Targeting NIL-Linking Cases. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM '22)*, October 17–21, 2022, Atlanta, GA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3511808.3557659>

¹The code is available at: <https://github.com/iurshina/NILK>.

²The dataset is distributed under CC BY-SA 3.0 licence.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CIKM '22, October 17–22, 2022, Atlanta, Georgia

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-XXXX-X/18/06...\$15.00

<https://doi.org/10.1145/3511808.3557659>

1 INTRODUCTION

Entity Linking matches a reference of a named entity in a text (known as a mention) with the corresponding entity in a specific knowledge base. It has gained increasing attention in recent years due to its usefulness in a wide range of downstream tasks from Question Answering [2] to Text Mining [12]. In a real-world scenario, the given knowledge base is often incomplete, which leads to cases where the mention in the text does not have a matching entity in the knowledge base. These mentions are called *NIL-mentions*, and the task of dealing with NIL-mentions is NIL-linking. In the NIL-linking task, we distinguish two sub-tasks: *NIL-detection* and *NIL-disambiguation*. NIL-detection determines if a mention is an NIL-mention, and if its corresponding entity exists in the knowledge base. If not, NIL-disambiguation will distinguish between NIL-mentions by determining which of them refers to the same out-of-knowledge-base entity. Detection and disambiguation of NIL-mentions allow people to constructively address the problem that corresponding entities do not exist in the target knowledge base by adding detected out-of-knowledge-base entities. NIL-disambiguation, in particular, ensures that the missing entity is only added once and all mentions that refer to it are connected accordingly.

Some Entity Linking datasets [4, 3] annotate all NIL-mentions into one special class, which can be used for NIL-detection task. Despite their successful efforts, NIL-disambiguation task is unexplored: these datasets do not distinguish between NIL-mentions and do not further annotate if some of NIL-mentions belong to the same out-of-knowledge-base entity. Due to the lack of available data source, NIL-linking and NIL-disambiguation tasks are often not considered and left for future work in recent Neural Entity Linking approaches [15, 6, 8, 10]. Consequently, current Neural Entity Linking approaches suffer from an inevitable restriction in practice as they can not provide detailed information about NIL-mentions for downstream applications. Therefore, it is necessary to move NIL-mentions' annotation forward from a single universal class to individual out-of-knowledge-base entities' IDs.

This paper presents NILK, a NIL-linking dataset dealing with both NIL-detection and NIL-disambiguation. Two editions of WikiData dump, WikiData 2017 and WikiData 2021 are used to annotate NIL-mentions accurately. We consider WikiData 2017 as our knowledge base and entities which do not exist in WikiData 2017 but exist in WikiData 2021 as out-of-knowledge-base entities. Afterwards, we annotate linked mentions whose relevant entities are in

the knowledge base and NIL-mentions whose relevant entities are out-of-knowledge base entities in English Wikipedia text with constructed new IDs. By comparing the changes of Wikidata entities from two WikiData dumps at different time points, we simulate the natural emergence of new entities in WikiData. An alternative approach would be to remove some of the entities from a specific dump randomly. However, using two dumps allows us to have a more realistic setting where the added entities are either less frequent or belong to named entities that appeared later (e.g., new books or people who gained popularity recently).

We conduct a comprehensive analysis of dataset characteristics, which can motivate promising improvement directions. Additionally, we provide the code used for the construction so it can be applied to different WikiData dumps or modified to incorporate additional information.

2 RELATED WORK

A variety of datasets have been proposed for Entity Linking in recent years [13, 9]. In general, most datasets adopt WikiData or Wikipedia as the knowledge base.

There are multiple datasets for Entity Linking with WikiData that do not contain NIL-mentions. We can cite, ISTEEX-1000 [1], which is a manually annotated dataset created to evaluate OpenTapioca [1] that contains one thousand author affiliation strings. Another example is T-REx [5] which was constructed automatically based on Wikipedia abstracts. And, Kensho Derived Wikimedia Dataset³ is a large-scale dataset that contains text extracted from Wikipedia with mentions annotated based on Wikipedia links and combined with a corresponding WikiData id. TweekiData [7] is a large dataset annotated automatically. It contains tweets text with mentions linked to WikiData. These datasets, however, only annotate linked mentions and do not consider NIL-mentions.

In general, only a few datasets deal with NIL-mentions. CLEF HIPE 2020 [3] is a challenge that includes a new dataset to evaluate named entity recognition and linking to historical newspapers in French, German and English. The main objectives of this challenge were (i) to test the robustness of existing approaches on non-standard inputs, (ii) to evaluate the performance of NE processing on historical texts, and (ii) to foster semantic indexing of historical documents. It contains 18962 mentions, and around 27% of them are labeled as NIL-mentions. However, the dataset is annotated only for NIL-entity detection tasks and not NIL-entity linking. A second example is Microposts2016 challenge [11]. This challenge task comprised automatic linking and classification of entities appearing in different event streams of English tweets. It contains 10025 mentions, and around 26% of them are labeled as NIL-mentions. Because of the tweets' writing style, the context for mentions in Microposts2016 contains high amounts of spoken language and slang.

The datasets that adopt Wikipedia as the target knowledge base could be converted to Entity Linking with WikiData as the knowledge base by matching Wikipedia pages with WikiData entities. For example, TACKBP-2010 [4] contains NIL-mentions with equivalence clusters making it suitable for the task of NIL-disambiguation after converting Wikipedia entities by mapping relevant Wikipedia

pages to associate WikiData id. Unfortunately, this dataset⁴ is not free, and its size is limited (1074 and 1020 annotated mention-entity pairs in the train and test sets, respectively).

3 DATASET STRUCTURE AND CONSTRUCTION

Our ultimate goal is to construct a NIL-linking dataset for both NIL-detection and NIL-disambiguation tasks simultaneously, which requires necessary information, including linked mentions with their corresponding entity IDs in the knowledge base, NIL-mentions with corresponding new assigned IDs for out-of-knowledge-base entities, and the context for mentions. Several knowledge bases such as WikiData, YAGO, and DBPedia have been used in different Entity Linking datasets. In this paper, we choose WikiData[14] as the knowledge base for three reasons: 1) It is one of the most popular knowledge bases nowadays, with more than 20000 active users⁵. 2) It has a broad coverage over different kinds of entities, and thus it is suitable for a universal Entity Linking setting. 3) It has a tight connection with Wikipedia text, reducing the annotation workload. To help researchers acquire additional information they might desire, we also provide Wikipedia page ID for each mention which can be used as an index to download the original Wikipedia page content. In the following sections, we first introduce the data collection process and then describe the process of annotating linked mentions and NIL-mentions. Finally, we give an overview of the constructed dataset.

3.1 Data Collection

We collect two types of data to construct the dataset: two snapshots of WikiData (from 2017 and 2021) and Wikipedia pages from 2017.

- (1) The snapshot of WikiData_2017 is used as the target knowledge base in NILK.
- (2) The English Wikipedia text in 2017 is used as the source text where mentions exist.
- (3) The trickiest part is to find out-of-knowledge-base entities. We assume that the most updated WikiData dump can serve as a supplementary material to older versions as emerging entities are continuously added into WikiData. Hence, we use WikiData_2021 as the real-world entity supplement to WikiData_2017.

3.2 Dataset Construction

As shown in Figure 1, our construction process contains five stages:

Stage 1: Out-of-Knowledge-Base Entity Detection. As mentioned before, we consider WikiData_2017 as the knowledge base and WikiData_2021 as its real-world supplement. Hence, we compare the two WikiData dumps and obtain the set of entities that are in WikiData_2021 but not in WikiData_2017 as out-of-knowledge-base entities. To further distill abstract entities, we check that entities we chose have the "instance of" relation and do not have a "subclass" relation in Wikidata. This allows us to limit the number of abstract notions and obtain more concrete mentions.⁶ Since we

⁴<https://catalog.ldc.upenn.edu/LDC2018T16>

⁵<https://www.wikidata.org/wiki/Wikidata:Statistics>

⁶Of course, this approach has limitations because WikiData relations contain mistakes and are incomplete.

³<https://www.kaggle.com/datasets/kenshoresearch/kensho-derived-wikimedia-data>

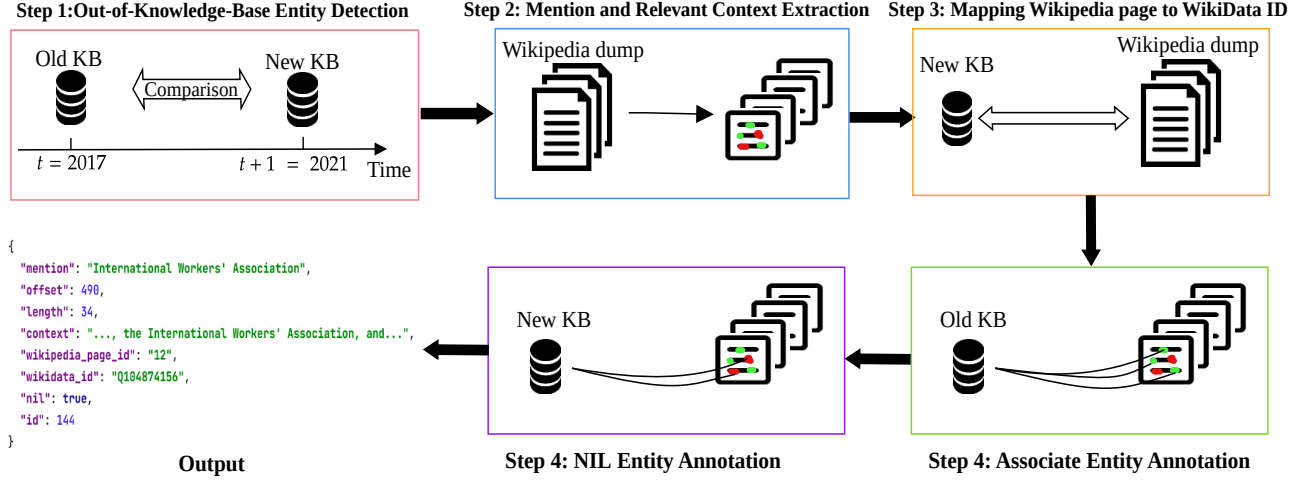


Figure 1: The pipeline of NILK dataset construction.

use English Wikipedia as the source of the text, we only take into account entities that have a link to an English Wikipedia page.

Stage 2: Mention and Relevant Context Extraction. In the second step, we detect meaningful mentions in Wikipedia text and extract suitable context for each mention. We only consider tokens with hyperlinks in Wikipedia text as mentions because most of the meaningful words in Wikipedia with real-world significance are linked to corresponding Wikipedia pages. To provide context for NIL-models, we extract at most 500 tokens before and after the mention. In this step, we only use Wikipedia pages that belong to the main/article namespace⁷.

Stage 3: Mapping Wikipedia page to WikiData ID. From both WikiData dumps, we obtain the mapping from the linked Wikipedia page title to WikiData ID. The hyperlink of each mention in Wikipedia, which we obtained in Stage 2, redirects to its corresponding entity’s main page, which helps us index relevant WikiData ID.

Stage 4: Associate Entity Annotation. We iterate through all mentions detected in Stage 2 and enrich their representation with an ID from WikiData_2017 if we have a WikiData_2017 ID mapping for this mention. To achieve this, we use the mapping obtained in Stage 3. We get WikiData ID from the mapping by Wikipedia link title we have on Wikipedia pages. Wikipedia links contain both the surface form of the mention and the title of the page it refers to. For the mapping, we use the title of the page, but the mention itself is the surface form of the link.⁸ This preserves the natural ambiguity of the mentions and gives us the reliable link for the mapping.

Stage 5: NIL Entity Annotation. Similar to Stage 4, we iterate through mentions, and if their linked Wikipedia pages’ mappings belong to the out-of-knowledge-base entities list on Stage 1, we mark them as NIL-mentions and add the WikiData_2021 ID to it. The WikiData_2021 ID serves as the new group ID for NIL-mentions and can be used for NIL-disambiguation.

3.3 Dataset Description

In this section, we perform a comprehensive analysis of NILK to provide a deeper understanding of the dataset and the task of NIL-linking.

Dataset Structure. Our final dataset consists of two parts: 1) A WikiData_2017 snapshot serves as the knowledge base for NIL-linking. 2) A Mentions file contains extracted mentions from Wikipedia text in 2017 with relevant context and corresponding entities’ information. More concretely, the Mentions file contains a collection of JSON-objects with the following fields: mention, offset, length, context, wikipedia page ID, wikidata ID, "is NIL" flag, and mention id.

As can be seen in Figure 2, "mention" field corresponds to the mention’s tokens span, "offset" refers to the number of characters from the beginning of the context field to the beginning of the mention, "length" is mention’s length, "context" extracts at most 500 characters before and after the mention (we ignore those mentions that have fewer than 10 tokens of context before and after them), "wikidata ID" is the ID from WikiData_2021 or WikiData_2017 dump and "is NIL" signifies whether the mention has a link in WikiData 2017 (the knowledge base) or NIL-linked. The "ID" field is a unique ID that we generated for each mention in the construction process.

Mentions file contains all mentions that were linked with the WikiData_2021⁹. So, with respect to WikiData_2017 as the knowledge base, it contains both linked mentions and NIL-mentions. As Mentions file stores for each of the mention a WikiData ID from WikiData_2021 or WikiData_2017, we can tell which of the mentions refer to the same entity by using these IDs.

Dataset Size. Table 1 provides statistical information on the dataset. About 6 % of entities are NIL-entities and about 1% of mentions are NIL-mentions. However, in absolute numbers it is a significant amount of data for the tasks of NIL-detection and NIL-disambiguation.

⁷<https://en.wikipedia.org/wiki/Wikipedia%3aNamespace>

⁸[https://en.wikipedia.org/wiki/Help:Link#Wikilinks_\(internal_links\)](https://en.wikipedia.org/wiki/Help:Link#Wikilinks_(internal_links))

⁹For linked mentions, their corresponding entities have the same WikiData ID in WikiData_2017 and WikiData_2021.

```
{
  "mention": "William Godwin",
  "offset": 456,
  "length": 14,
  "context": "... such as William Godwin and Wilhelm...",
  "wikidata_id": "Q188569",
  "nil": false,
  "id": 8
}
```

Figure 2: A mention instance in NILK**Table 1: NILK statistics**

Type	Count
Linked entities	4,228,124
Out-of-knowledge-base entities	352,765
Linked mentions	106,028,997
NIL-mentions	1,652,484

Table 2 demonstrates the distribution of the number of mentions per entity for NIL cases. About half of the entities have more than one mention, which provides substantial data for the task of NIL-disambiguation.

Table 2: Number of mentions per NIL-entity

Number of mentions	Number of NIL-entities
1	158,524
2 to 5	129,272
6 to 100	63,822
101 to 1000	1090
1000+	57

3.4 Dataset Split

We provided the dataset split into approximately 80/10/10 ratio. We make sure that the splits do not intersect in terms of entities following the "zero-shot" manner, that is, Validation and Test splits do not contain any entities from the Train split or from each other. Statistics on the splits can be found in Table 3. We use L-Entities to refer to linked entities.

In addition to that, to ensure a meaningful distribution of number of mentions per entity between splits, the splits were made within several "buckets" separately. The "buckets" with entities were sampled proportionately based on how many mentions they have. The proportion for "buckets" we used correspond to the ones shown in Table 2. Additionally, the splitting was done separately for NIL and linked entities.

3.5 Comparison with Other Datasets

Table 4 shows how NILK compares in size with two other datasets for Entity Linking based on WikiData that contain NIL-mentions. We omit TACKBP-2010 as it is not free and its knowledge base is

Table 3: Dataset split ("L" stands for "linked").

Split	L-Entities	NIL-Entities	L-mentions	NIL-mentions
Train	3,382,497	282,210	85,052,764	1,327,039
Val	422,812	35,276	10,525,107	162,948
Test	422,815	35,279	10,451,126	162,497

Wikipedia. As shown in the table, NILK is significantly larger than CLEF HIPE 2020 and Microposts2016 in terms of NIL-mentions. Besides, NILK is only one to provide annotation for the NIL-disambiguation task. We hope the large-scale NILK dataset could drive the development of NIL-disambiguation task.

Table 4: Dataset comparison between multiple datasets. The second column #mentions refers to the number of linked mentions and NIL-mentions. The third column #NIL-mentions refers to the number of NIL-mentions.

Name	# mentions	# NIL-mentions
CLEF HIPE 2020	18,962	4579
Microposts2016	10,025	2660
NILK	107,681,481	1,652,484

4 CONCLUSION AND FUTURE WORK

In this paper, we present NILK, a large-scale NIL-linking dataset. The main advantage of NILK is that it deals with the underexplored NIL-disambiguation annotation task and provides a significant amount of data for the task of NIL-mentions disambiguation without manual labeling.

For future work, several directions can be explored. Firstly, the approach we took to detect which NIL-mentions correspond to the same out-of-knowledge-base entity is not entirely realistic. Naturally, the latest WikiData dump is still incomplete, meaning that we can not successfully annotate all NIL-mentions with it. However, a better approach would be desirable, which does not rely on human annotations.

Secondly, adjusting various dataset splits could be considered more challenging in experimental settings. For example, we might ensure not only that there is no intersection in entities but also that entities are not linked neighbors in the knowledge base between splits. We would set a minimum for the number of hops between them to achieve this.

Thirdly, to improve the completeness of out-of-knowledge-base entity coverage, our dataset could be extended to a dynamic NIL-linking dataset by automatically applying the described algorithm to the up-to-date WikiData dump each year.

REFERENCES

- [1] Antonin Delpuch. 2020. Opentapioca: lightweight entity linking for wikidata. In *Proceedings of the 1st Wikidata Workshop (Wikidata 2020) co-located with 19th International Semantic Web Conference (ISWC 2020), Virtual Conference, November 2-6, 2020* (CEUR Workshop Proceedings). Lucie-Aimée Kaffee, Oana Tifrea-Marcuska, Elena Simperl, and Denny Vrandečić, (Eds.) Vol. 2773. CEUR-WS.org. <http://ceur-ws.org/Vol-2773/paper-02.pdf>.

- [2] Mohnish Dubey, Debayan Banerjee, Debanjan Chaudhuri, and Jens Lehmann. 2018. Earl: joint entity and relation linking for question answering over knowledge graphs. In *International Semantic Web Conference*. Springer, 108–126.
- [3] Maud Ehrmann, Matteo Romanello, Alex Flückiger, and S. Clematide. 2020. Extended overview of clef hi pe 2020: named entity processing on historical newspapers. In *CLEF*.
- [4] Joe Ellis, Xuansong Li, Kira Griffitt, Stephanie M Strassel, and Jonathan Wright. 2012. Linguistic resources for 2013 knowledge base population evaluations. In *TAC*.
- [5] Hady Elsahar, Pavlos Vougiouklis, Arslan Remaci, Christophe Gravier, Jonathon Hare, Frederique Laforest, and Elena Simperl. 2018. T-REx: A Large Scale Alignment of Natural Language with Knowledge Base Triples. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. Nicoletta Calzolari (Conference chair) et al., (Eds.) European Language Resources Association (ELRA), Miyazaki, Japan, (May 2018). ISBN: 979-10-95546-00-9.
- [6] Octavian-Eugen Ganea and Thomas Hofmann. 2017. Deep joint entity disambiguation with local neural attention. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Copenhagen, Denmark, (Sept. 2017), 2619–2629. <https://aclanthology.org/D17-1277>.
- [7] Bahareh Harandizadeh and Sameer Singh. 2020. Tweeki: linking named entities on Twitter to a knowledge graph. In *Proceedings of the Sixth Workshop on Noisy User-generated Text (W-NUT 2020)*. Association for Computational Linguistics, Online, (Nov. 2020), 222–231. <https://aclanthology.org/2020.wnut-1.29>.
- [8] Feng Hou, Ruili Wang, Jun He, and Yi Zhou. 2020. Improving entity linking through semantic reinforced entity embeddings. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 6843–6848.
- [9] Hideaki Joko, Faegheh Hasibi, Krisztian Balog, and Arjen P de Vries. 2021. Conversational entity linking: problem definition and datasets. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2390–2397.
- [10] Phong Le and Ivan Titov. 2019. Boosting entity linking performance by leveraging unlabeled documents. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Florence, Italy, (July 2019), 1935–1945. <https://aclanthology.org/P19-1187>.
- [11] G Rizzo, MGJ van Erp, J Plu, and R Troncy. 2016. Making sense of microposts (# microposts2016) named entity recognition and linking (neel) challenge. In *Proceedings of the 6th Workshop on Making Sense of Microposts*, 50–59.
- [12] Özge Sevgili, Artem Shelmanov, Mikhail Arkhipov, Alexander Panchenko, and Chris Biemann. 2022. Neural entity linking: a survey of models based on deep learning. *Semantic Web*, 13, 3, (Apr. 2022), 527–570. Mehwish Alam et al., (Eds.) <https://doi.org/10.3233/2Fsw-222986>.
- [13] Marieke Van Erp, Pablo Mendes, Heiko Paulheim, Filip Ilievski, Julien Plu, Giuseppe Rizzo, and Jörg Waitelonis. 2016. Evaluating entity linking: an analysis of current benchmark datasets and a roadmap for doing a better job. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*, 4373–4379.
- [14] Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: a free collaborative knowledgebase. *Commun. ACM*, 57, 10, (Sept. 2014), 78–85. doi: 10.1145/2629489.
- [15] Ledell Wu, Fabio Petroni, Martin Josifoski, Sebastian Riedel, and Luke Zettlemoyer. 2020. Scalable zero-shot entity linking with dense entity retrieval. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Online, (Nov. 2020), 6397–6407. <https://aclanthology.org/2020.emnlp-main.519>.